of RLK (from above) based on the composition of the optimal trade list. This will also result in a more appropriate allocation of dollars across assets and across stocks. The portfolio will be more efficient. Furthermore, since the byproduct of the optimization will include both the new targeted portfolio and the underlying instructions and trading schedule to achieve that portfolio there will be no ambiguities between the investment objectives and trading goals. Traders will be provided with the underlying execution strategy (directly from the optimizer) to be used to transact those shares. Portfolio manager and trader goals will finally be aligned!

In the remainder of the chapter, we provide the necessary background and quantitative framework to assist portfolio managers and traders properly align investment objectives and trading goals. This results in a single best execution trading strategy for the specific investment decision. We expand the Markowitz efficient trading frontier to include transaction costs and show there are various cost-adjusted frontiers but only one efficient trading frontier. The chapter concludes with an introduction to the necessary mathematics and optimization process to develop multi-period portfolio optimizers incorporating transaction cost analysis.

Some of the highlights of the chapter include:

- Unification of the investment and trading decisions resulting in consistency across all phases of the investment cycle and providing a true best execution process.
- There exist multiple cost-adjusted efficient investment frontiers but a single optimal trading strategy.
- The Sharpe ratio determines the appropriate level of risk aversion for trade schedule optimization.
- Evidence that a naïve VWAP strategy is often an inefficient execution strategy because it may lead to lower levels of investor utility and suboptimal ex-post portfolios.
- Evidence that a passive VWAP strategy and an aggressive execution strategy may result in identical levels of investor utility.
- Portfolio optimization framework that properly incorporates market impact and timing risk estimates. This leads to an improved best execution frontier and optimal ex-post portfolios.
- An approach to determine whether a suboptimal Markowitz portfolio exists resulting in a more efficient and pareto optimal portfolio after trading costs. For example, it may be possible for an ex-ante suboptimal portfolio to have higher risk-return characteristics (and be more optimal) than the originally optimal portfolio ex-post.